Kayla McFarlane

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EDUCATION

Carnegie Mellon University **Pittsburgh, PA**
Master of Science in Robotics *August 2027*
Relevant Coursework: Machine Learning, Math Fundamentals for Robotics, Embodied AI Safety, Robot Learning

Bachelor of Science in Electrical and Computer Engineering (*Software Systems Concentration)* *May 2025*
Relevant Coursework: Computer Systems, Computer Security, Computer Vision, Applied Cryptography, Data-Focused Python, Mobile Web Design & Development, Web Application Development, Human-Robot Interaction, Ethics and Robotics, Mobile Robots

RESEARCH EXPERIENCE

Graduate Research Assistant *May 2025 – Present*
SAFE (Safe Agile Function Everywhere) Benchmark Project, CMU AirLab

- Built a benchmark framework that evaluates how autonomous navigation policies behave under uncertainty, scoring behavior from observed trajectories alone with no access to system internals or dynamics models
- Formalized safety through adopter-configurable regions, trajectory-level risk metrics, an outcome taxonomy, and CVaR-based validation criteria that account for finite-sample uncertainty
- Implemented the framework across two simulation backends, including Isaac Sim with PX4 software-in-the-loop, placing published planners and learned policies behind a common interface so they run under identical conditions
- Integrating the framework into AirStack as a portable evaluation layer, demonstrating that it wraps around an existing autonomy stack without modification to the policies under test

Graduate Research Assistant *May 2026 – Present*
Strike vs Guard (SVG), CMU AirLab

- Contributing to a multi-institute project developing an autonomous multi-agent aerial swarm for adversarial engagement, in which a defending team performs coordinated area denial against an intruding team using control barrier function-based control
- Developed the protected teleoperation component, constraining human operator input so that manual control of a drone preserves the system's safety guarantees, implemented as a branch of AirStack

Undergraduate Research Assistant *Mar 2024 – May 2025*
SAFE Project, CMU AirLab

- Contributed to the SAFE (Safe Agile Function Everywhere) project, focused on enabling agile autonomous drone navigation in unknown, dynamic, and cluttered environments with degraded perception
- Conducted extensive literature reviews on aerial robotics safety frameworks, analyzing approaches from collision avoidance to perception-aware navigation
- Collaborated with graduate students to systematically evaluate state-of-the-art methods for agile flight, identifying key limitations and opportunities for improvement
- Participated in weekly research discussions to develop novel approaches combining neural Control Barrier Functions (CBFs), Hamilton-Jacobi reachability, and Vision-Language Models
- Set up and configured 3D simulation environments, in Isaac Gym, for testing perception-conditioned Neural CBFs, maintaining comprehensive documentation
- Contributed to research on Perception-Aware Neural Control Barrier Functions for adaptive safety in degraded visual environments

Undergraduate Research Assistant *Jan 2023 – Feb 2024*
Wildfire Project, CMU AirLab

- Contributed to the Wildfire Project, aimed at providing real-time situational awareness to firefighters during wildfires
- Developed components of FIReVision, a photorealistic wildfire simulation platform for testing drone navigation, object detection, and multimodal sensing in degraded environments.
- Designed and implemented an automated synthetic-data pipeline that generated **15,000+** RGB, thermal, depth, and segmentation images for perception-model training.
- Built modular Python tools for drone control, object detection, data collection, and pipeline validation across simulated wildfire environments.
- Created realistic smoke effects in Unreal Engine using FluidNinja to support perception testing under degraded visibility.
- Presented FIReVision research at CMU's Meeting of the Minds Undergraduate Research Symposium and contributed technical demos for stakeholder and grant presentations.

Human-Robot Interaction Research Project *Nov 2023 – Dec 2023*
Human-Robot Interaction, Carnegie Mellon University
Exploring Accent Variability, Phrase Difficulty, and Speaker Distance in Robot-Enabled Speech Recognition

- Evaluated Google Speech Recognition integrated with Pepper robot across diverse accents (American, Chinese, Guyanese, Brazilian) to assess ASR performance in classroom settings

- Developed Python modules for real-time transcription and analyzed performance using Translation Error Rate (TER) metrics
- Identified critical limitations in ASR accuracy for accented speech and distance variability, providing recommendations for improving accessibility in educational settings
- Project code adopted by AI Maker Space for demonstrating speech recognition challenges

Undergraduate Research Observer *Nov 2024*
The MindfulNest Project, CMU CREATE Lab
- Observed real-world interactions between neurodivergent children and **MindfulNest**, a socially assistive emotional regulation robot
- Documented and **analyzed behavioral patterns** to identify design insights around engagement, comfort, and emotional expression
- Supported data collection and contributed insights to refine the pilot study methodology

PUBLICATIONS

Manuscripts in Preparation
- **K. McFarlane** and S. Scherer. "SAFE: A Black-Box Safety Evaluation Benchmark for Autonomous Mobile Robots." In preparation for IEEE Robotics and Automation Letters (RA-L)

Workshop Papers
- Navin Sriram Ravie, Andrew Jong, **Kayla McFarlane**, Jay Patrikar, Bijo Sebastian, Sebastian Scherer. "Don't Fool Me Twice: Adapting to Failures in the Wild with Multi-Modal Reasoning." *IROS 2025 Workshop on The Art of Robustness: Surviving Failures in Robotics*, October 2025

PRESENTATIONS & POSTERS
- "SAFE: A Black-Box Safety Evaluation Benchmark for Autonomous Mobile Robots." Poster, Midwest Robotics Workshop (MWRW), Toyota Technological Institute at Chicago, June 2026.
- "FIReVision: A Photorealistic Thermal IR Simulator for Wildfire Robotics." Poster, Meeting of the Minds Undergraduate Research Symposium, Carnegie Mellon University, May 2024.

AWARDS & HONORS
Rales Fellow, Carnegie Mellon University Aug *2025 – Aug 2027*
- Awarded a prestigious full-tuition fellowship to one of 34 underrepresented students in STEM from 400+ applications, providing mentorship, career coaching, and professional development opportunities to cultivate future STEM leaders
- Selected by Senior Vice Provost for Academic Initiatives to represent Carnegie Mellon University in officially welcoming NVIDIA CEO Jensen Huang alongside the Dean of SCS and Dean of Engineering ahead of the 2026 commencement ceremony, discussing my research on autonomous robot safety evaluation (May 2026)

2nd Place, IEEE RAS Standards Video Prize Competition, IROS 2026 *Oct 2026*
- Recognized for a robotics standards video and awarded $4,000 in travel support for ICRA 2027 in Seoul, South Korea.

ROSCon Global 2026 Scholarship Recipient | Toronto, Canada *Sep 2026*
- Awarded full conference registration, accommodation, and travel support to attend the fifteenth ROSCon Global, a three-day developer conference on the Robot Operating System.

RimA Summer School, FZI Research Center and Karlsruhe Institute of Technology *Aug 2026*
- One of selected participants from 450 global applicants for a fully funded summer school on walking robots in domestic environments, covering 3D mapping, machine learning, ROS2, and behavior trees with hands-on deployment on Boston Dynamics Spot and Unitree Go2

ETH Zurich Robotics Summer School, Selected Participant *Jun 2025*
- One of 50 students globally selected from 292 applicants for a fully funded summer program on robotic optimization, hands-on UGV programming, search and rescue challenges, and field deployment in real-world environments

Frank J. Marshall Undergraduate Scholarship Recipient, Carnegie Mellon University *Aug 2023 – May 2025*

INDUSTRY EXPERIENCE
Chief Innovation Officer & Lead Project Manager *May 2026 – Present*
AquaMind 2035 Program, M-PowerHouse of Greater Pittsburgh
- Instruct a selective cohort of high school students in robotics foundations, drone technologies, AI systems, and engineering design through hands-on project based learning
- Support student development in workforce readiness, technical communication, and STEAM career pathways
- Lead program design and execution across four structured learning tiers, ensuring alignment with grant milestones and partner deliverables

Software Engineering Intern *Jun 2024 – Aug 2024*
DigITs Program, Memorial Sloan Kettering Cancer Center (MSK)
- Developed AI chatbot interface using React and TypeScript for patient portal serving **170,000+** users annually
- Streamlined LLM fine-tuning process for **50+** developer teams, improving AI integration across departments
- Studied full-stack architecture of MyMSK's Self-Scheduling system (**React** frontend, **C#/.NET** backend, **SQL** Server).
- Led development of MSK SkinMonitor, winning 1st place in intern hackathon for AI-powered health tracking app
- Practiced Agile methodologies through daily stand-ups and two-week sprint cycles

TECHNICAL PROJECTS

INFERNOT – AI-Powered Wildfire Defense System *Nov 2025*
CMU Hack-A-StartUp Competition
- Collaborated with 4-member team to develop automated wildfire protection system using AI sensors and robotics
- Led robotics integration strategy for automated deployment system using thermal imaging, computer vision, and air-quality sensing
- Contributed to technical feasibility analysis and prototype development planning
- Placed **4th** out of 15 teams in university-wide entrepreneurship competition

Nightingale – Voice Cleaner and Transcriber *Oct 2025*
Wireless Innovation Hackathon for Accessibility
- Developed web-based assistive technology to enhance voice clarity and communication for individuals with vocal cord disorders in noisy environments
- Implemented audio processing pipeline using Sox for noise reduction and Whisper AI for real-time speech transcription
- Designed future wearable device concept (Nightingale Clip) for portable, real-time voice assistance
- Collaborated with interdisciplinary team and engaged directly with disability community stakeholders

Study Bearbot – Interactive Desktop Companion for Focus and Stress Relief *Jan 2025 – May 2025*
Electrical and Computer Engineering Capstone Project, Carnegie Mellon University
- Collaborated with a partner to design and build a **multisensory desktop robot** that supports student focus and stress reduction
- Led firmware development on the **Arduino Nano RP2040**, integrating tactile button inputs, toggle switches, rotary encoder, and OLED display to control LED lighting, ultrasonic diffuser, and audio playback with sub-500ms responsiveness
- Built a React-based web app for real-time, two-way synchronization with the robot, enabling remote control of sensory features (LED, aroma, audio), timer tracking, and dynamic motivational quote display through HTTP communication
- Achieved an average **7.8-hour** battery life and demonstrated **measurable reductions** in **user stress**
- Conducted an **8-user study** demonstrating reduced stress and increased focus, integrating survey data and interaction logs to inform design iterations

Up2 - Location-Based Social Networking Application *Nov 2024 – Dec 2024*
Web Application Development Course Project, Carnegie Mellon University
- Contributed to development of location-based social networking platform using React, Django, JavaScript, Bootstrap, HTML5 Geolocation API, and Google Maps API with real-time location tracking and privacy controls
- Engineered Friends Panel interface using React and Bootstrap for managing social connections with integrated backend data flows via Django
- Designed mobile-responsive About Page using Bootstrap's grid system for user onboarding
- Developed frontend components for status sharing and privacy toggles with JavaScript-driven interactivity synchronized to server-side configurations

Vivian: A Caribbean Voice Assistant *Feb 2024*
SheInnovates Hackathon 2024
- Developed voice assistant prototype for underrepresented English dialects using OpenAI Whisper
- Implemented speech-to-text system for Caribbean Patois using Python and custom API integrations
- Led technical design and development of speech processing pipeline
- Addressed challenges in training a potential AI model due to insufficient datasets for Caribbean accents, highlighting the need for further data collection and model fine-tuning for underrepresented English dialects

JADA House International Website Redesign *Oct 2022 – Dec 2022*
- Led team redesign of nonprofit website to improve accessibility and user engagement across devices
- Collaborated with founder to gather requirements and iterate on UI/UX designs based on user feedback
- Implemented responsive site using HTML, CSS, Bootstrap with Figma-to-code workflow
- Conducted user testing to validate improvements in usability and mobile responsiveness

VOLUNTEER

Conference Student Volunteer *Sep – Oct 2026*
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) | Pittsburgh, PA
- Assisted with attendee registration, check-in, and on-site conference logistics.

Conference Volunteer *Mar 2026*
National Society of Black Engineers (NSBE) Annual Convention | Baltimore, Maryland
- Represented the Rales Fellowship at Carnegie Mellon University booth, advising prospective graduate students on STEM and robotics programs
- Answered questions about funding pathways including Rales Fellowship and GEM Fellowship, and shared insights on navigating graduate school applications

Conference Volunteer *Jan 2026*
IEEE/SICE International Symposium on System Integration (SII 2026) | Cancun, Mexico
- Supported conference operations for international robotics and systems integration symposium
- Assisted with session coordination and attendee support

Outreach Committee Member *Oct 2024 – Dec 2024*
Society of Women Engineers (SWE), Carnegie Mellon University
- Participated in biweekly planning meetings for SWE's High School Day, contributing to event logistics and programming

- Helped coordinate and set up the event day activities and venues
- Introduced young women to engineering through hands-on experiences and provided mentorship during the event

SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, C#, HTML/CSS, SQL, MATLAB
Frameworks/Libraries: React, .NET Core, Node.js, Express.js, Django, Bootstrap
Tools: Git, GitHub, Linux, Docker, Figma, Isaac Gym/Sim/Lab, Gazebo, ROS2
Technologies: SQL Server, REST APIs, Agile/Scrum
Technical Areas: Full-Stack Development, Computer Vision, Machine Learning/AI, Embedded Systems, Real-time Systems, API Integration, Sensor Integration

EXTRACURRICULARS

Women in Electrical and Computer Engineering (WinECE), Society of Women Engineers (SWE), Carnegie AI Safety Initiative (CASI), Paths to AI Research (PAIR)